

Roll No.:

MID SEMESTER EXAMINATION MARCH – 2025

Name of the Program: B.Tech. Biotechnology
Name of the Course: Statistical Data Analysis with R
Time: 90 Minutes

Semester: VI
Course Code: TCS 471
Maximum Marks: 50

Note:

- i) Answer **all the questions** by choosing any one of the sub questions.
- ii) Each question carries 10 marks.

Q1	(10Marks)	
(a)	A consignment of 15 record players contains 4 defectives. The record players are selected at random, one by one, and examined. Those examined are not put back. What is the probability that the 9 th one examined is the last defective?	CO1
	OR	
(b)	What do you understand by Point Estimation? Define the following terms and give one example for each: (i) Consistent Statistic (ii) Unbiased Statistic (iii) Sufficient Statistic (iv) Efficiency	
Q2	(10Marks)	
(a)	(i) Define sampling distribution and standard error. Obtain standard error of mean when population is large. (ii) Find the standard error of a linear function of several variables. Deduce the standard error of the mean of n uncorrelated variables following the same distribution.	CO1
	OR	
(b)	A sample of 400 individuals is found to have a mean height of 67.47 inches. Can it be reasonably regarded as a sample from a large population with mean height of 67.39 inches and standard deviation 1.30inches?	
Q3	(10Marks)	
(a)	(i) State and explain the principle of maximum likelihood for estimation of population parameter. (ii) Describe the maximum likelihood method of estimation and discuss five of its optimal properties.	CO1
	OR	CO2
(b)	In answering a question on a multiple choice test a student either knows the answer or he guesses. Let p be the probability that he knows the answer and $1 - p$ the probability that he guesses. Assume that a student who guesses at the answer will be correct with probability $1/5$, where 5 is the number of multiple-choice alternatives. What is the conditional probability that a student knew the answer to a question given that he answered it correctly?	
Q4	(10Marks)	
(a)	Comment on the following with proper explanation: "The mean of a binomial distribution is 3 and variance is 4".	CO2
	OR	
(b)	Suppose that the number of telephone calls that an operator receives from 9:00 to 9:05 hours in a day follows a Poisson distribution with mean 3. Find the probability that (i) the operator will receive no calls in that time interval tomorrow, (ii) In the next three days the operator will receive a total of 1 call in that time interval.	
Q5	(10 Marks)	

(a)	(i) In a distribution exactly normal, 7% of the items are under 35 and 89% are under 63. What are the mean and standard deviation of the distribution? (ii) Of a large group of men, 5% are under 60 inches in height and 40% are between 60 and 65 inches. Assuming a normal distribution, find the mean height and standard deviation.	CO2
	OR	CO3
(b)	How do you add a new column to a data frame in R?	